DART Installation Guide

This document provides detailed installation instructions for DART (Digital Asset Routing and Transformation) on macOS, Windows, and Linux systems.

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macOS Installation (DMG)

DART for macOS is distributed as a DMG file with an automated installer that requires **no admin permissions**.

Prerequisites

- **macOS 12** (Monterey) or later
- **Python 3.8+** must be installed
  - Download from: python.org/downloads
  - Or install via Homebrew: `brew install python`
  - Verify: `python3 --version`

Installation Steps

1. **Download the DMG**
   - Download `DART_vX.x.dmg` from your distribution source
2. **Open the DMG**
   - Double-click the downloaded DMG file
   - A new window opens showing the installer contents.

Note for Mac users who are UPDATING DART to a new version: You may need to remove or move your `~/DART/` folder to get it out of the way BEFORE the `Install DART.command` can run successfully!

3. **Run the Installer**
   - Double-click: `Install DART.command`

   Note for first-time users (unsigned app) and updates:

- If the installer won't open, macOS Gatekeeper is blocking it
- Right-click on `Install DART.command` → select "Open"
- If the app is blocked close the dialog then proceed to `System Settings...` and `Privacy & Security`
- The key is to look for a statement like "Install DART.command was blocked..." and there should be an `Open Anyway` button beside it. Click that button to allow the install.
- This is only needed once

4. **Follow the Installation**

   - A Terminal window opens with the installer
   - The installer checks for Python 3
   - Confirms installation location: `~/DART/`
   - Copies all files (takes a few seconds)
   - Creates a launcher on your Desktop
   - Press Enter when finished

5. **Eject the DMG**

   - Right-click the mounted DART volume → Eject
   - Or drag it to the Trash

Installation Location

DART is installed to: `~/DART/` (your home directory)

```
~/DART/
├── Launch DART.command   # Double-click to start DART
├── run.sh                # Root compatibility launcher
├── app.py                # Main application
├── scripts/              # Launcher and helper scripts
│   └── run.sh            # Launcher script
├── common_dg_utilities/  # Utility functions (included)
├── INSTALLATION.md       # This file
├── README.md            # Full documentation
└── ... (other files)
```

Desktop Shortcut

The installer creates: **Launch DART** on your Desktop

- Double-click this to launch DART at any time
- It's a symbolic link to `~/DART/Launch DART.command`

First Launch

1. **Start DART**

   - Double-click: `Launch DART` (on Desktop)
   - Or navigate to `~/DART/` and double-click: `Launch DART.command`

2. **Automatic Setup (first time only)**
    - A Terminal window opens
    - Python virtual environment is created in `~/DART/.venv/`
    - Dependencies install automatically (1-2 minutes)
    - Progress is shown in the Terminal
3. **DART Window Opens**
    - The DART application window appears when ready
    - The Terminal window stays open (don't close it!)
    - Minimize the Terminal if you want it out of the way
4. **Subsequent Launches**
    - Just double-click the launcher
    - DART starts immediately (no setup needed)
    - Dependencies are already installed

Why No Applications Folder?

Traditional approach issues:

- Many work computers restrict Applications folder access
- Requires admin permissions to install
- Users encounter permission errors

DART's solution:

- Installs to `~/DART/` in your home directory
- No admin permissions required
- Works on all systems, including managed/restricted computers
- Clean, predictable location

Windows Installation (ZIP)

DART for Windows is distributed as a ZIP archive.

Prerequisites

- **Windows 10** or **Windows 11**
- **Python 3.8+** must be installed
    - Download from: python.org/downloads
    - **During installation: Check "Add Python to PATH"** ✅
    - Verify: Open Command Prompt, type `python --version`

Installation Steps

1. **Download the ZIP**
    - Download `DART_v2.x_Windows.zip` from your distribution source

2. **Extract the Archive**
    - Right-click the ZIP file → Extract All...
    - Choose a convenient location (e.g., `C:\Users\YourName\DART\`)
    - Or extract to your Desktop
3. **Installation Location**
    - The extracted folder contains all DART files
    - No further installation needed

First Launch

1. **Open the DART folder**
    - Navigate to where you extracted the files
2. **Run the launcher**
    - Double-click: `run.bat`
3. **Automatic Setup (first time only)**
    - A Command Prompt window opens
    - Python virtual environment is created in `.venv\`
    - Dependencies install automatically (1-2 minutes)
    - Progress is shown in the console
4. **DART Window Opens**
    - The DART application window appears when ready
    - The Command Prompt window stays open (don't close it!)
    - Minimize the console if you want it out of the way
5. **Subsequent Launches**
    - Just double-click `run.bat`
    - DART starts immediately

Creating a Desktop Shortcut (Optional)

1. Right-click `run.bat` → Send to → Desktop (create shortcut)
2. Rename the shortcut to "DART"
3. Double-click the Desktop shortcut to launch DART

Linux Installation (Source)

DART runs from source on Linux systems.

Prerequisites

- **Python 3.8+**

- Most Linux distributions include Python 3
- Verify: `python3 --version`
- Install if needed:
  - Debian/Ubuntu: `sudo apt install python3 python3-venv`
  - Fedora/RHEL: `sudo dnf install python3`
  - Arch: `sudo pacman -S python`

Installation Steps

1. **Download or Clone**

```
# Clone from repository:
git clone https://github.com/Digital-Grinnell/DART.git
cd DART
```

2. **Install common-DG-utilities (required dependency)**

```
cd ..
git clone https://github.com/Digital-Grinnell/common-DG-utilities.git
cd DART
```

3. **Run DART**

```
./run.sh
```

4. **First Launch**
   - The script creates a Python virtual environment in `.venv/`
   - Installs all dependencies automatically
   - Launches the DART application
   - Subsequent runs skip the setup phase

Note for developers: The `python_requirements.txt` file includes `-e ../common-DG-utilities` which references the sibling repository. This line is automatically removed by the build scripts when creating DMG/ZIP distributions since those packages bundle the utilities directly.

Creating a Desktop Launcher (Optional)

Create `~/.local/share/applications/dart.desktop`:

```
[Desktop Entry]
Type=Application
Name=DART
Comment=Digital Asset Routing and Transformation
Exec=/path/to/DART/scripts/run.sh
```

```
Icon=utilities-terminal
Terminal=false
Categories=Utility;
```

Update `/path/to/DART/` with your actual path.

Prerequisites

Python 3.8 or Later

DART requires Python 3.8 or later. To verify your Python version:

```
# macOS/Linux
python3 --version

# Windows
python --version
```

Installation:

- **macOS:**
  - Download: python.org/downloads
  - Or via Homebrew: `brew install python`
- **Windows:**
  - Download: python.org/downloads
  - **Important:** Check "Add Python to PATH" during installation
- **Linux:**
  - Usually pre-installed
  - If needed: `sudo apt install python3` (Debian/Ubuntu)

Python Dependencies

The following Python packages are installed automatically on first launch:

- **flet 0.25.2** — UI framework
- **flet-desktop 0.25.2** — Desktop-specific features
- **cryptography** — Encrypted settings storage
- **azure-storage-blob 12.19.0+** — Azure Blob Storage integration
- **Pillow** — Image processing
- **PyMuPDF (fitz)** — PDF processing
- **pandas** — Data manipulation

You don't need to install these manually — the `run.sh` / `scripts/run.sh` / `scripts/run.bat` launchers handle everything.

Troubleshooting

macOS Issues

"Install DART.command can't be opened"

Cause: macOS Gatekeeper blocking unsigned app

Solution:

1. Right-click on `Install DART.command`
2. Select "Open" from the menu
3. Click "Open" in the security dialog
4. Only needed once

"Python 3 not found" error

Cause: Python not installed or not in PATH

Solution:

```
# Install Python via Homebrew
brew install python

# Or download from python.org
# Then verify:
python3 --version
```

Installer says "DART is already installed"

Cause: Previous DART installation exists

Solution:

- Type `y` to overwrite the old installation
- Or type `n` to cancel and manually remove `~/DART/` first

DART window doesn't appear

Cause: Terminal window was closed during setup

Solution:

1. Open Terminal
2. Navigate: `cd ~/DART`
3. Run manually: `bash run.sh`
4. Watch for error messages

Windows Issues

"python is not recognized"

Cause: Python not installed or not in PATH

Solution:

1. Install Python from python.org
2. **During installation: Check "Add Python to PATH"**
3. Restart Command Prompt
4. Verify: `python --version`

Dependencies fail to install

Cause: Network issues or pip outdated

Solution:

```
# Upgrade pip
python -m pip install --upgrade pip

# Then re-run DART
scripts\run.bat
```

Virtual environment errors

Cause: Corrupt virtual environment

Solution:

```
# Delete the virtual environment
rmdir /s .venv

# Re-run launcher (recreates .venv)
scripts\run.bat
```

Linux Issues

`./run.sh`: Permission denied

Cause: Script not executable

Solution:

```
chmod +x run.sh scripts/run.sh
./run.sh
```

"No module named 'tkinter'"

Cause: tkinter not installed (required by flet)

Solution:

```
# Debian/Ubuntu
sudo apt install python3-tk

# Fedora/RHEL
sudo dnf install python3-tkinter
```

Common Issues (All Platforms)

DART closes immediately after launch

Cause: Error during startup

Solution:

1. Look for error messages in the Terminal/Console window
2. Check log files in `~/DART-data/logfiles/` or `{working_folder}/logfiles/`
3. Common causes:
    - Missing Python dependencies (re-run installer)
    - Corrupted settings file (delete `~/DART-data/persistent.json`)

"common_dg_utilities not found" error

Cause: DMG/ZIP was missing the utilities folder

Solution:

- This should not happen with properly built installers v2.2+
- The build scripts automatically bundle common_dg_utilities
- For developers: Ensure `../common-DG-utilities/` exists when building

"not a valid editable requirement" error

Cause: Using source repository with missing sibling common-DG-utilities repo

Solution (for developers):

```
# Clone the required utilities repository
cd ..
git clone https://github.com/Digital-Grinnell/common-DG-utilities.git
cd DART
```

Note: DMG/ZIP installers automatically remove this dependency since utilities are bundled directly.

Uninstallation

macOS

1. **Delete the application folder:**

```
rm -rf ~/DART
```

2. **Delete user data (optional):**

```
rm -rf ~/DART-data
```

3. **Remove Desktop shortcut:**
   - Drag `Launch DART` from Desktop to Trash

Windows

1. **Delete the DART folder:**
   - Navigate to where you extracted DART
   - Delete the entire folder
2. **Delete user data (optional):**
   - Navigate to: `C:\Users\YourName\DART-data\`
   - Delete the folder
3. **Remove Desktop shortcut (if created):**
   - Delete the DART shortcut from Desktop

Linux

```
# Delete DART folder
rm -rf ~/path/to/DART

# Delete user data (optional)
rm -rf ~/DART-data

# Remove desktop launcher (if created)
rm ~/.local/share/applications/dart.desktop
```

Support

Documentation

- **README.md** — Full application documentation
- **QUICKSTART.md** — Quick reference guide
- **Function Help** — Built-in help for each function (Help Mode toggle in app)

Getting Help

- **Repository:** https://github.com/Digital-Grinnell/DART
- **Issues:** Report bugs on GitHub Issues
- **Questions:** Use GitHub Discussions

Log Files

DART maintains detailed log files for troubleshooting:

- **Startup logs:** `~/DART-data/logfiles/`
- **Operational logs:** `{working_folder}/logfiles/` (set in app)
- **Format:** `dart_YYYYMMDD_HHMMSS.log`

Include relevant log excerpts when reporting issues.

System Requirements

Minimum

- **OS:** macOS 12+, Windows 10+, or modern Linux
- **Python:** 3.8 or later
- **Memory:** 2 GB RAM
- **Disk:** 500 MB free space
- **Display:** 1280x720 or higher

Recommended

- **OS:** macOS 13+, Windows 11, or latest LTS Linux
- **Python:** 3.10 or later
- **Memory:** 4 GB RAM
- **Disk:** 1 GB free space
- **Display:** 1920x1080 or higher

Security Note

DART stores sensitive settings (like Azure connection strings) using encrypted storage:

- **Encryption key:** `~/DART-data/encryption_key`
- **Settings file:** `~/DART-data/persistent.json` (sensitive fields encrypted)
- **Working settings:** `{working_folder}/dart_settings.json` (per-project)

Keep your encryption key secure. If lost, you'll need to re-enter encrypted settings.

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Documentation: https://github.com/Digital-Grinnell/DART